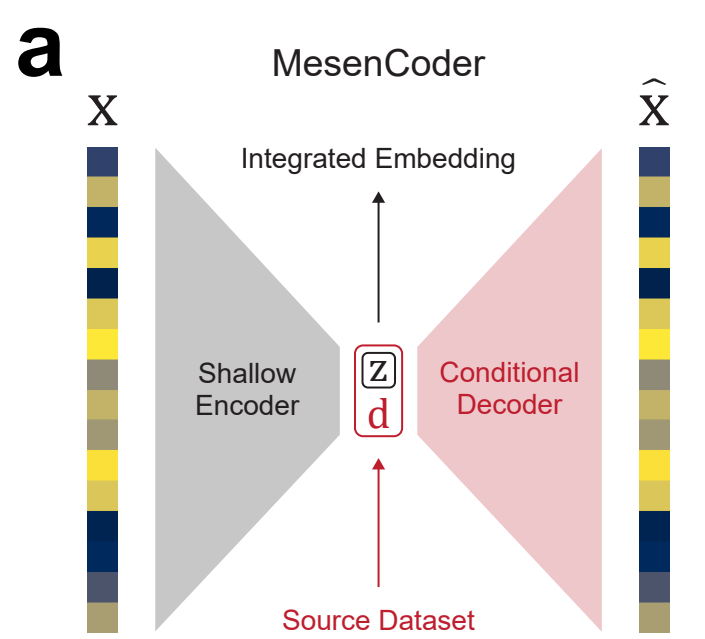
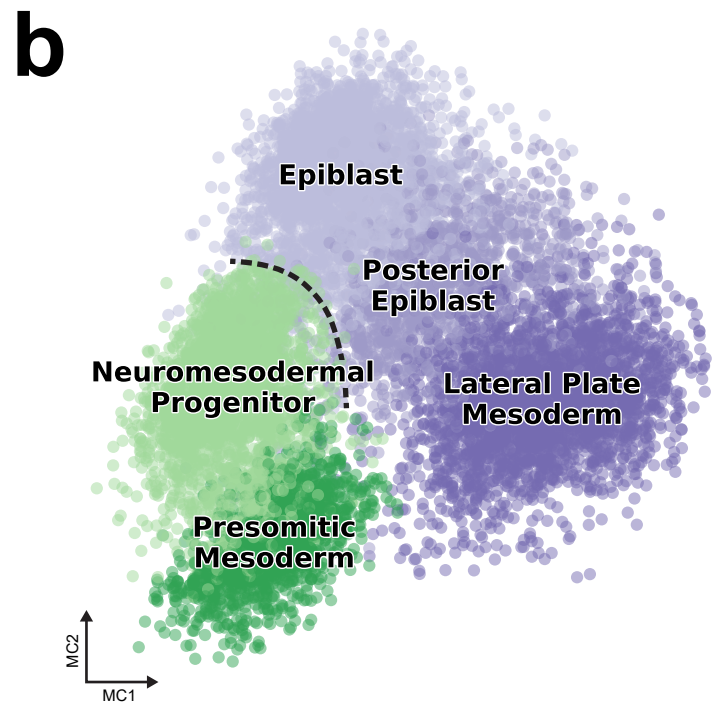
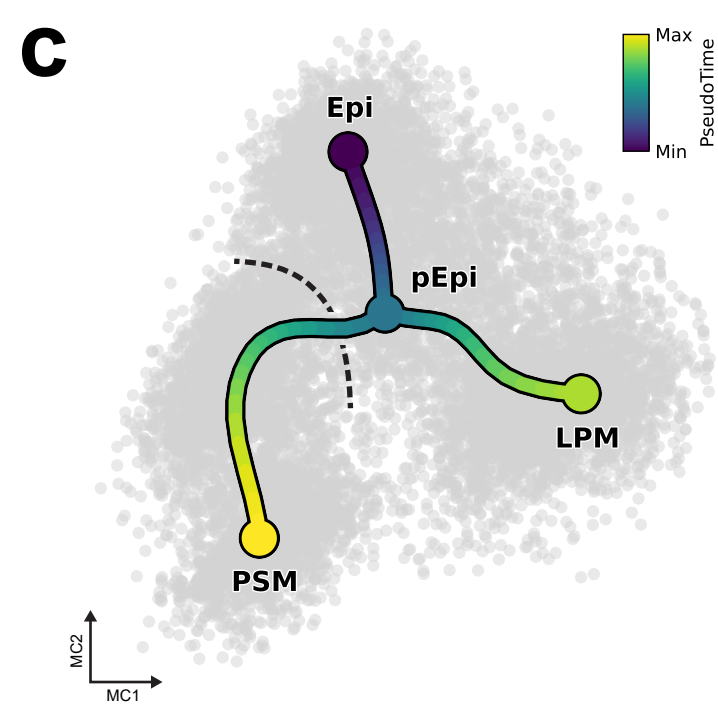
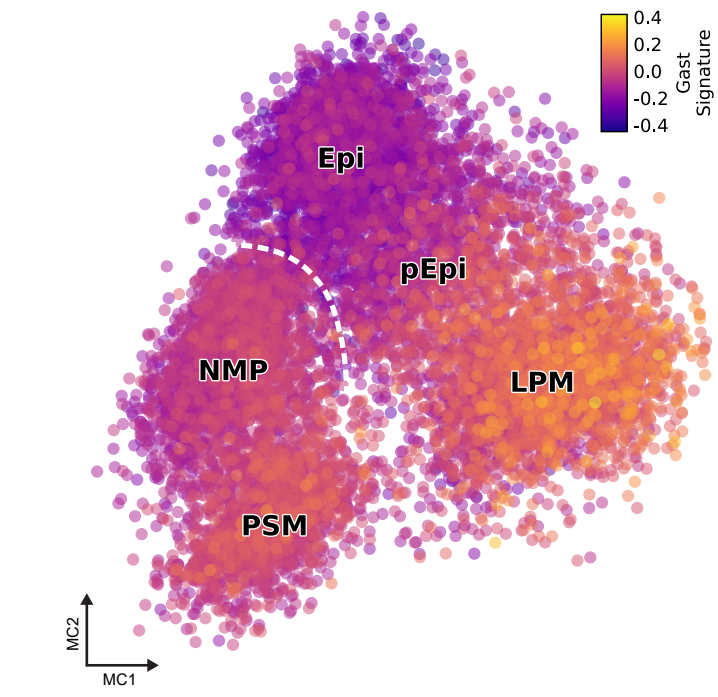
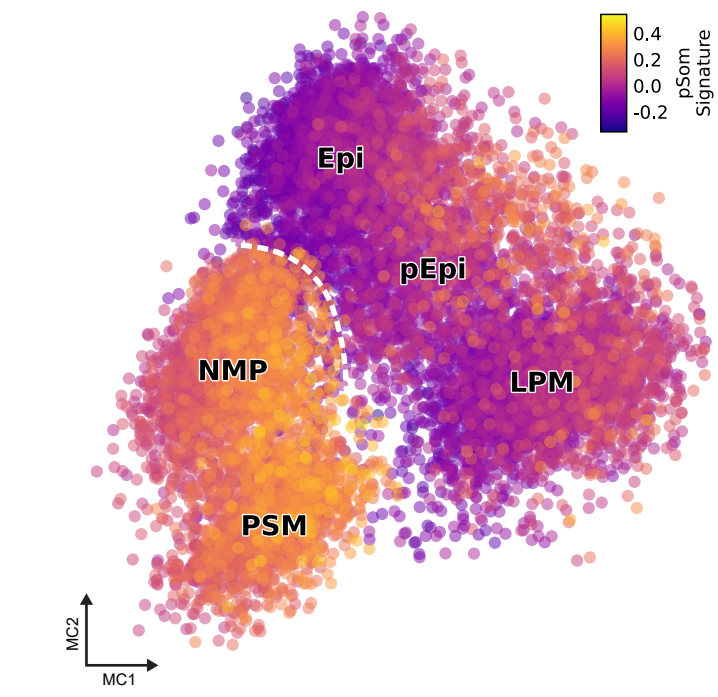
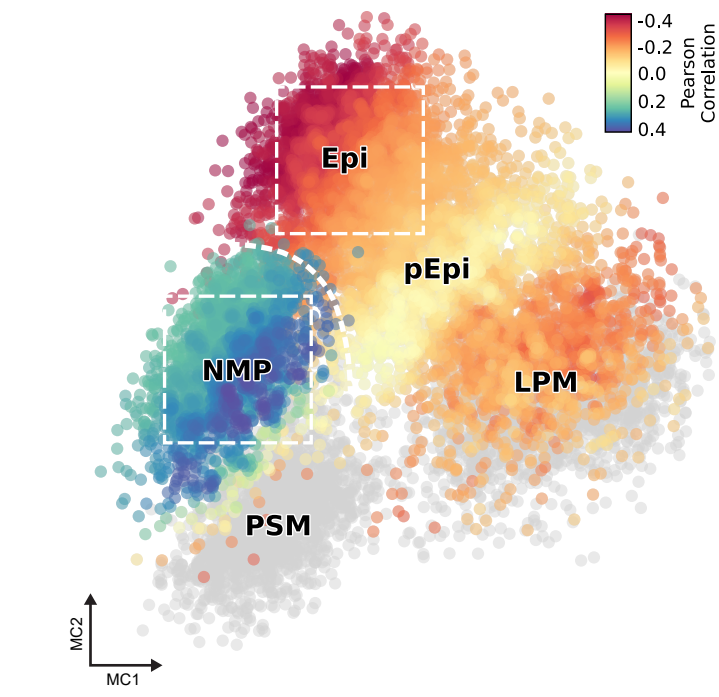

a
MesenCoder
X
$\hat{X}$
Integrated Embedding
Shallow Encoder
z
d
Conditional Decoder
Source Dataset

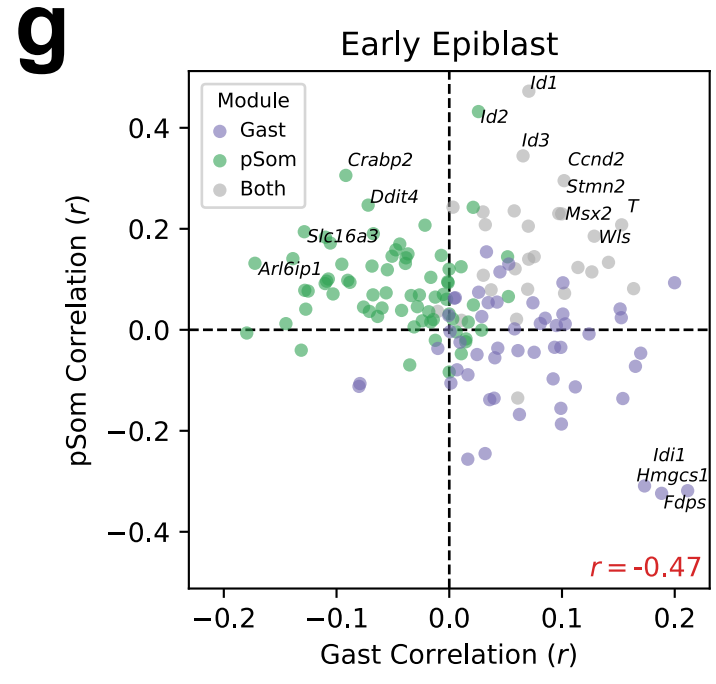

b
Epiblast
Posterior Epiblast
Neuromesodermal Progenitor
Lateral Plate Mesoderm
Presomitic Mesoderm
MC2
MC1

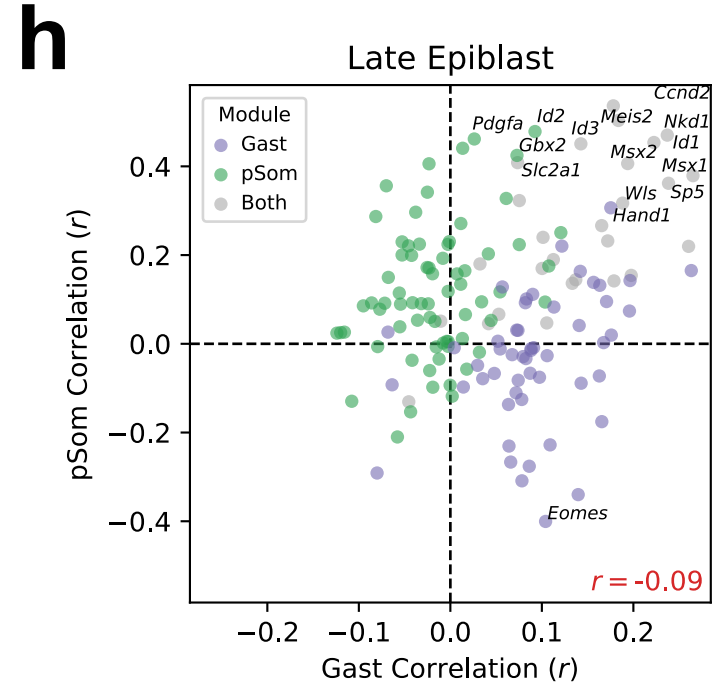

c
Max
Min
PseudoTime
Epi
pEpi
LPM
PSM
MC2
MC1

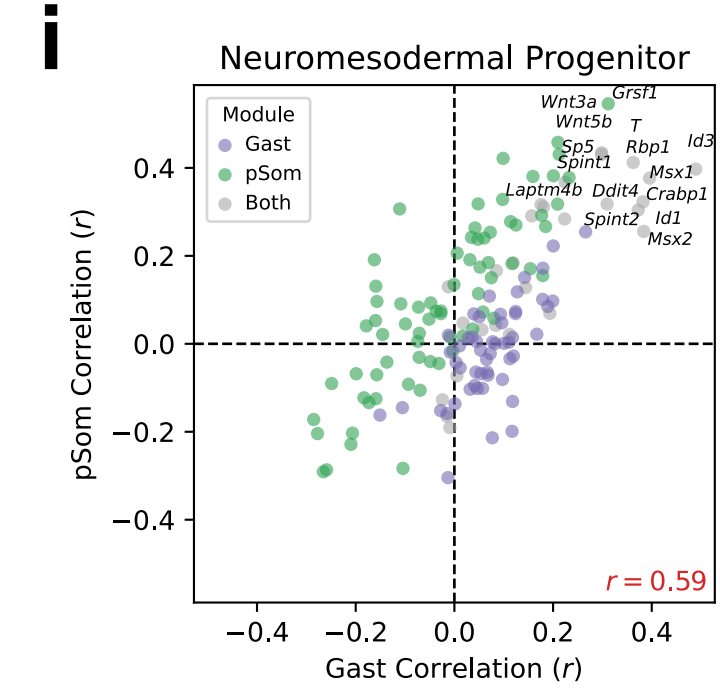

d
Gastrulation Signature
0.4
0.2
0.0
-0.2
-0.4
Gast Signature
Epi
pEpi
NMP
LPM
PSM
MC2
MC1


e
Somitogenesis Signature
0.4
0.2
0.0
-0.2
pSom Signature
Epi
pEpi
NMP
LPM
PSM
MC2
MC1


f
Module Correlation
-0.4
-0.2
0.0
0.2
0.4
Pearson Correlation
Epi
pEpi
NMP
LPM
PSM
MC2
MC1


g
Early Epiblast
Module
Gast
pSom
Both
Id1
Id2
Id3
Crabp2
Ccnd2
Stmn2
Ddit4
Msx2
T
Slc16a3
Wls
Arl6ip1
Idi1
Hmgcs1
Fdps
r = -0.47
pSom Correlation (r)
Gast Correlation (r)


h
Late Epiblast
Module
Gast
pSom
Both
Ccnd2
Pdgfa
Id2
Meis2
Nkd1
Id3
Gbx2
Id1
Msx2
Msx1
Slc2a1
Wls
Sp5
Hand1
Eomes
r = -0.09
pSom Correlation (r)
Gast Correlation (r)


i
Neuromesodermal Progenitor
Module
Gast
pSom
Both
Wnt3a
Grsf1
Wnt5b
T
Sp5
Rbp1
Id3
Spint1
Msx1
Laptm4b
Ddit4
Crabp1
Spint2
Id1
Msx2
r = 0.59
pSom Correlation (r)
Gast Correlation (r)